MA 2101 : Analysis I

August 31, 2025

NOTATIONS

- $\mathbb{R}$, $\mathbb{N}$ denote the set of real numbers and the set of natural numbers respectively.
- sup denotes the supremum or the least upper bound of a set.
- inf denotes the infimum or the greatest lower bound of a set.

Problem 1.

(a) State the completeness property/axiom for the set of real numbers in the context of upper bounds.

(b) Suppose $x, y \in \mathbb{R}$ such that $x > 0$. Use the completeness axiom to show the existence of a natural number n such that $nx > y$.

[2+3=5]

Problem 2.
Show that $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$. [3]

Problem 3.
Let A, B be non-empty subsets of $\mathbb{R}$ such that $a \leq b$ for all $a \in A$ and all $b \in B$. Note that A is bounded above, while B is bounded below. Then show that

$$\sup A \leq \inf B.$$

[3]

Problem 4.
Show that the greatest lower bound of the set $\{n + \frac{1}{m} : m, n \in \mathbb{N}\}$ is 1. [4]

IISER Kolkata Quiz-II

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Maximum Marks: 15

Problem 1.
Let a_n be a sequence defined as follows :

$$a_1 = 2,\ a_{n+1} = \frac{1}{2}(a_n + \frac{2}{a_n}),\ n \geq 2.$$

Show that the sequence a_n converges to $\sqrt{2}$. [5]

Problem 2.
Suppose $a_1 \geq a_2 \geq a_3 \geq ... \geq 0$. Then prove that $\sum_{n=1}^{\infty} a_n$ converges if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges. [4]

Problem 3.
Check for convergence/divergence of the following series of non-negative terms :
(a) $\sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}}$. (b) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$ [3+3=6]